

HOSPITAL MANAGEMENT SYSTEM

Complete Project Report

Python-Based Command-Line Hospital Service and Billing Application

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Prepared according to the supplied Hospital Management System source code and project documentation. The report includes command-line execution evidence, billing output, error-handling evidence, and a full-code visual appendix.

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1. Abstract

Hospital Management System is a Python-based command-line application designed to manage a small set of hospital services through a menu-driven terminal workflow. The system collects patient information and allows the user to select doctor appointment, medical tests, hospital room, and food services before generating a consolidated bill.

The project demonstrates first-semester programming concepts including input/output, variables, functions, conditional statements, loops, validation, arithmetic operations, and modular menu-based control. The application is intentionally terminal-based and does not require third-party packages.

2. Introduction

The application provides a simple workflow from patient registration to service selection and bill generation. Patient name, age and phone number are collected first. The main menu then routes the user to individual service modules. Selected charges are accumulated and shown in a final bill.

3. Problem Statement

- Accept and validate basic patient information.
- Provide separate options for doctor appointment, medical tests, hospital room and food services.
- Allow the user to select services through a command-line menu.
- Calculate and display a consolidated payment amount.
- Prevent invalid menu selections and validate important numeric inputs.
- Run completely from a terminal without a GUI-based setup.

4. Objectives

- Develop a first-semester Python application with a complete command-line workflow.
- Demonstrate functions, conditions, loops, input/output and arithmetic operations.
- Organize hospital services into understandable menu-driven modules.
- Validate age, phone number and service-selection inputs.
- Generate a clear itemized final bill.

5. Project Overview

The program starts by collecting patient details and then presents a main menu. The user can choose one service at a time, return to the main menu, and finally generate the bill. The service modules use predefined charge values and the final amount is calculated from the selected services.

6. Technologies and Requirements

Item	Project Choice
Programming language	Python 3.x
Application type	Command-line / terminal application
Core concepts	Functions, conditions, loops, input/output, arithmetic
External packages	None required
Execution environment	Terminal / Command Prompt / Shell

The project documentation specifies direct execution through the Python interpreter. The source is designed for a standard Python 3.x environment.

7. Project Structure

The project is centered on the main Python source file and supporting project documentation.

File / Component	Purpose
cs project hospital management.py	Complete Python command-line implementation.
README / project documentation	Project description, setup, usage and execution guidance.
Service functions	Doctor, tests, room, food and billing workflow.
Validation logic	Checks phone, age, menu and service inputs.

8. Functional Requirements

ID	Requirement	Implementation
FR-01	Patient Registration	Collect name, age and phone number.
FR-02	Phone Validation	Accept a 10-digit phone number.
FR-03	Doctor Service	Select doctor/appointment and add charge.
FR-04	Medical Tests	Select a test and add its charge.
FR-05	Room Service	Select room and duration; calculate charge.
FR-06	Food Service	Select food option and add charge.
FR-07	Billing	Display itemized charges and total payment.
FR-08	Menu Handling	Reject invalid menu/service selections.

9. Non-Functional Requirements

Category	Requirement
Usability	Simple menu-driven command-line interaction.
Portability	Runs in a standard Python 3.x terminal environment.
Dependencies	No third-party Python packages are required.
Reliability	Input validation provides messages for invalid entries.
Maintainability	Service responsibilities are separated into functions.
Performance	Operations are small and suitable for first-semester scale.
Execution Constraint	Runs directly from a terminal without GUI setup.
Data Persistence	Patient/service data exists only during the current execution.

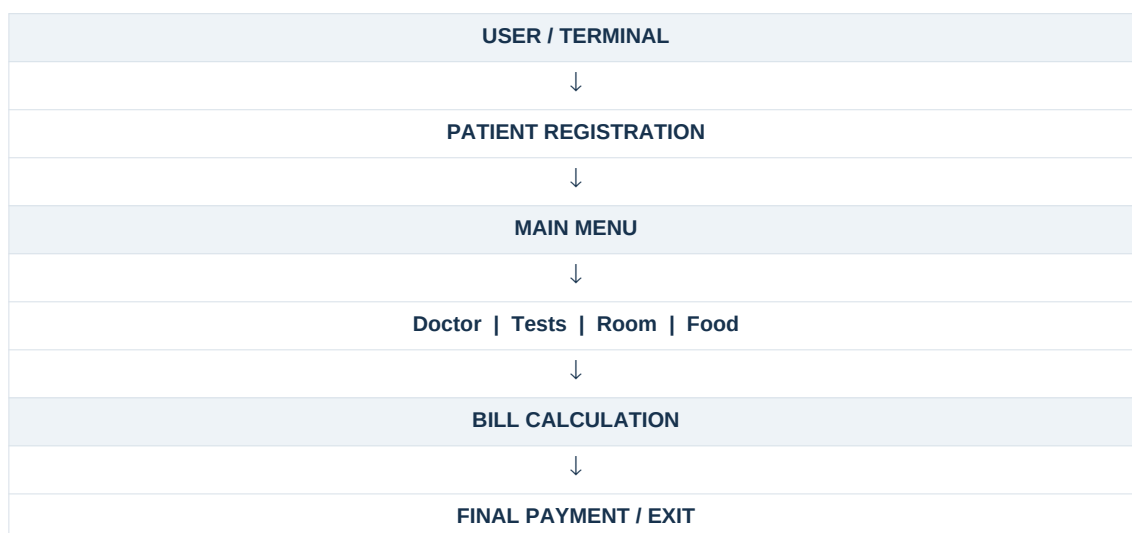
10. System Architecture

The system follows a simple layered command-line flow. The user interacts with the terminal, the main program controls navigation, service functions calculate individual charges, and the billing logic combines the selected amounts into the final total.

Layer / Component	Responsibility
Command-Line Interface	Reads input and displays output.
Main Menu / Control	Routes the user to service modules.
Patient Registration	Stores patient name, age and phone.
Service Modules	Doctor, tests, room and food selection.
Billing Module	Adds service charges and displays final payment.
Validation	Checks numeric ranges and invalid selections.

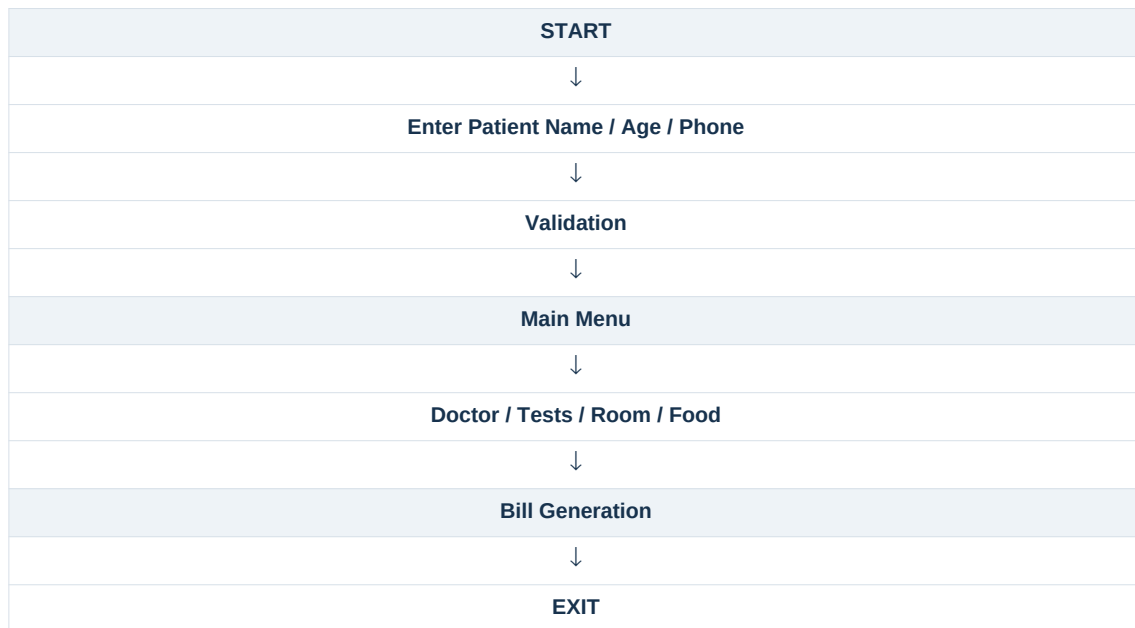
11. Design Diagrams

High-level flow: Patient Registration → Main Menu → Service Selection → Service Charges → Final Bill → Exit



11. Design Diagrams – Workflow and Service Flow

Workflow: START → Enter Patient Details → Validate Inputs → Main Menu → Select Service → Calculate Charge → Return to Menu → Generate Final Bill → Exit.



12. Modules and Service Logic

Module	Main responsibility
Patient Registration	Collect and validate patient details.
Doctor Appointment	Display/select doctor-related service and charge.
Medical Tests	Select test and add test charge.
Hospital Room	Select room and duration; calculate room cost.
Food Services	Select food option and add food charge.
Billing	Display each selected charge and calculate total.
Exit	Terminate the command-line application cleanly.

13. Algorithms and Billing Logic

Patient validation: Read age → convert to integer safely → check permitted range → request again if invalid. Read phone → check numeric 10-digit format → continue only when valid.

Service selection: Display main menu → read choice → route to selected service → validate service-specific input → return to main menu.

Billing: Total Payment = Doctor Charges + Medical Test Charges + Room Charges + Food Charges.

Example: ■1000 + ■700 + ■2000 + ■150 = ■3850.

13. Algorithms and Billing Logic – Detailed Steps

- Initialize patient and service-charge variables.
- Collect patient name, age and phone number.
- Validate age and phone number before continuing.
- Display the main service menu.
- For each selected service, collect the required option and calculate its charge.
- Return to the main menu after each service.
- When billing is selected, print each charge separately.
- Add all service charges to obtain the total payment.
- Display the total and provide an exit option.

14. Error and Boundary Handling

Case	Expected handling
Invalid age	Display validation message and request a valid numeric age.
Invalid phone	Reject non-10-digit input and request again.
Invalid main-menu choice	Display an invalid-choice message.
Invalid doctor selection	Reject unsupported option and request valid selection.
Invalid test selection	Reject unsupported option and request valid selection.
Invalid room days	Reject invalid/non-positive duration.
Invalid food selection	Reject unsupported option and request valid selection.

The report includes separate visual evidence for these error-handling cases. The revised local execution used reusable integer-validation logic so non-numeric input does not terminate the program with an uncaught `ValueError`.

15. Command-Line Execution

The application is intended to run directly from a terminal. Typical execution is:

```
python "cs project hospital management.py"
```

The terminal workflow can be followed from patient registration through the service menus and final bill.

15. Command-Line Execution – Visual Evidence

```
●●● Step 1 – Program Startup and Patient Details

CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 1. Patient registration and initial command-line input.

```
●●● Step 2 – Main Menu

CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 2. Main service menu.

```
●●● Step 3 – Doctor Appointment

CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
DOCTOR APPOINTMENT

1. General Medicine - Rs. 500
2. Cardiology - Rs. 1000
3. Orthopedics - Rs. 700
4. Dermatology - Rs. 600

Enter doctor choice: Cardiology selected

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 3. Doctor appointment/service selection.

16. Testing and Execution Evidence

```
Step 4 - Medical Tests
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
MEDICAL TESTS
1. Blood Test - Rs. 300
2. X-Ray - Rs. 500
3. ECG - Rs. 400
4. MRI - Rs. 2500
5. Done

Enter test choice: Blood Test added
Enter test choice: ECG added
Enter test choice: Medical test selection completed

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 4. Medical test selection and charge flow.

```
Step 5 - Hospital Room
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
HOSPITAL ROOM
1. General Ward - Rs. 1000 per day
2. Semi-Private Room - Rs. 2000 per day
3. Private Room - Rs. 3500 per day

Enter room choice: Enter number of days: General Ward selected

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 5. Hospital room selection and duration.

```
Step 6 - Food Services
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
FOOD SERVICES
1. Breakfast - Rs. 150
2. Lunch - Rs. 250
3. Dinner - Rs. 250
4. Juice - Rs. 100
5. Done

Enter food choice: Breakfast added
Enter food choice: Food selection completed

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 6. Food service selection and charge.

16. Testing and Final Billing Evidence

The following evidence shows the service workflow reaching the final bill and then returning to the menu/exit stage.

```
Step 7 - Generate Bill
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
DOCTOR APPOINTMENT
1. General Medicine - Rs. 500
2. Cardiology - Rs. 1000
3. Dermatology - Rs. 700
4. Neurology - Rs. 800
5. Generate Bill
6. Exit

Enter doctor choice: Cardiology selected

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
MEDICAL TESTS
1. Blood Test - Rs. 200
2. X-Ray - Rs. 300
3. ECG - Rs. 400
4. MRI - Rs. 700
5. Bone
6. Exit

Enter test choice: Blood Test added
Enter test choice: ECG added
Enter test choice: MRI test selection completed

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
HOSPITAL ROOM
1. General Ward - Rs. 1000 per day
2. Semi Private Room - Rs. 1500 per day
3. Private Room - Rs. 2000 per day
6. Exit

Enter room choice: Enter number of days: General Ward selected

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
FOOD SERVICES
1. Breakfast - Rs. 150
2. Lunch - Rs. 200
3. Dinner - Rs. 250
4. Snacks - Rs. 100
5. Desserts
6. Exit

Enter food choice: Breakfast added
Enter food choice: Food selection completed

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
HOSPITAL BILL

Patient Name: Radhe
Patient Age: 20
Phone Number: 9876543210

Doctor Charges: Rs. 1000
Medical Test Charges: Rs. 700
Room Charges: Rs. 2000
Food Charges: Rs. 150

TOTAL PAYMENT: Rs. 3850

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 7. Complete run showing itemized final bill.

FINAL BILL CALCULATION

Doctor + Medical Tests + Room + Food = Total Payment

```
6. Exit

Enter your choice:
HOSPITAL BILL

Patient Name: Radhe
Patient Age: 20
Phone Number: 9876543210

Doctor Charges: Rs. 1000
Medical Test Charges: Rs. 700
Room Charges: Rs. 2000
Food Charges: Rs. 150

TOTAL PAYMENT: Rs. 3850

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit
```

Figure 8. Final bill calculation: $\blacksquare 1000 + \blacksquare 700 + \blacksquare 2000 + \blacksquare 150 = \blacksquare 3850$.

```
Step 8 - Clean Exit
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 9. Exit message after completion.

16. Testing – Error Handling Evidence

```
●●● Error Handling 1 – Invalid Age
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Invalid input! Please enter a number.
Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 10. Invalid age input handling.

```
●●● Error Handling 2 – Invalid Phone Number
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Invalid phone number!
Please enter exactly 10 digits.
Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 11. Invalid phone-number handling.

```
●●● Error Handling 3 – Invalid Main Menu Choice
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice: Invalid input! Please enter a number.
Enter your choice: Invalid input! Please enter a valid value.
Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!
```

Figure 12. Invalid main-menu selection.

```
●●● Error Handling 4 – Invalid Doctor Choice
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU

1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
DOCTOR APPOINTMENT

1. General Medicine - Rs. 500
2. Cardiology - Rs. 1000
3. Orthopedics - Rs. 700
4. Dermatology - Rs. 600

Enter doctor choice: Invalid input! Please enter a number.
Enter doctor choice: Invalid input! Please enter a valid value.
Enter doctor choice: Invalid input! Please enter a valid value.
Enter doctor choice:
```

Figure 13. Invalid doctor selection.

16. Testing – Additional Error Handling Evidence

```

Error Handling 5 – Invalid Test Choice
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
MEDICAL TESTS
1. Blood Test - Rs. 300
2. X-Ray - Rs. 500
3. ECG - Rs. 400
4. MRI - Rs. 2500
5. Done

Enter test choice: Invalid input! Please enter a valid value.
Enter test choice: Medical test selection completed

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!

```

Figure 14. Invalid medical-test selection.

```

Error Handling 6 – Invalid Room / Days
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
HOSPITAL ROOM
1. General Ward - Rs. 1000 per day
2. Semi-Private Room - Rs. 2000 per day
3. Private Room - Rs. 3500 per day

Enter room choice: Invalid input! Please enter a valid value.
Enter room choice: Enter number of days: Invalid input! Please enter a valid value.
Enter number of days: General Ward selected

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!

```

Figure 15. Invalid room-duration input.

```

Error Handling 7 – Invalid Food Choice
CITY CARE HOSPITAL
HOSPITAL MANAGEMENT SYSTEM

Enter patient name: Enter patient age: Enter 10-digit phone number: Phone number accepted

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
FOOD SERVICES
1. Breakfast - Rs. 150
2. Lunch - Rs. 250
3. Dinner - Rs. 250
4. Juice - Rs. 100
5. Done

Enter food choice: Invalid input! Please enter a number.
Enter food choice: Invalid input! Please enter a valid value.
Enter food choice: Food selection completed

MAIN MENU
1. Doctor Appointment
2. Medical Tests
3. Hospital Room
4. Food Services
5. Generate Bill
6. Exit

Enter your choice:
Thank you for visiting City Care Hospital!
Have a nice day!

```

Figure 16. Invalid food selection.

17. Complexity Analysis

Operation	Complexity	Reason
Patient input	O(1)	Fixed number of patient fields.
Menu selection	O(1)	Fixed menu options.
Service charge calculation	O(1)	Fixed option/charge lookup logic.
Bill calculation	O(1)	Fixed number of charge categories.

18. Strengths of the Project

- Clear command-line workflow suitable for a first-semester project.
- Separate service functions make the logic easier to explain.
- Itemized billing makes the final result easy to verify.
- Input validation is demonstrated through dedicated error cases.
- No third-party packages are required.
- Includes complete terminal execution evidence.

19. Limitations

- Command-line interface only; no graphical interface.
- Data is not permanently stored in a database or file.
- Service options and prices are predefined.
- The project is designed for a small academic workflow rather than hospital-scale deployment.

20. Future Scope

- Add database storage for patients, appointments and bills.
- Add authentication and role-based access.
- Add timestamps and searchable patient records.
- Add more hospital departments and services.
- Add automated testing and stronger validation.
- Add a graphical or web interface in a future version.

21. Results

The implemented system demonstrates a complete terminal-based hospital service workflow. It accepts patient details, routes the user through service modules, calculates individual charges, produces an itemized final bill, demonstrates invalid-input handling, and exits through the command-line interface.

22. Conclusion

Hospital Management System is a compact Python project that combines practical hospital-service billing with first-semester programming concepts. The project demonstrates structured functions, conditional logic, loops, input/output, arithmetic operations, validation and command-line execution. The final billing evidence confirms that the selected service charges are combined into a single total payment.

23. References

No.	Reference
1	Supplied Hospital Management System Python source file.
2	Supplied project README / documentation.
3	Python documentation for command-line execution and basic language features.
4	Project repository / source as supplied for the academic project.

Appendix A. Source Code Excerpts

The following excerpts represent the main structure of the supplied/revised command-line implementation. The complete source is preserved separately; this appendix focuses on the functions and billing flow most relevant to the report.

Patient validation pattern:

```
try: age = int(input('Enter patient age: ')) except ValueError: print('Invalid input. Please enter a number.')
```

Service charge accumulation:

```
doctor_charges = 1000 medical_test_charges = 700 room_charges = 2000 food_charges = 150 total_payment =  
doctor_charges + medical_test_charges + room_charges + food_charges
```

Bill display concept:

```
print('Doctor Charges:', doctor_charges) print('Medical Test Charges:', medical_test_charges) print('Room  
Charges:', room_charges) print('Food Charges:', food_charges) print('TOTAL PAYMENT:', total_payment)
```

The report distinguishes the original repository behavior from the revised numeric-input validation enhancement used for the execution evidence. The full-code visual on the next page is included as submission evidence.

Appendix B. Visual Evidence – Full Source Code

Hospital Management System – Complete Revised Source Code

```
001 print("CITY CARE HOSPITAL")
002 print("HOSPITAL MANAGEMENT SYSTEM")
003 print()
004
005 def get_integer(prompt, min_value=None, max_value=None):
006     while True:
007         try:
008             value = int(input(prompt))
009             if min_value is not None and value < min_value:
010                 print("Invalid input! Please enter a valid value.")
011                 continue
012             if max_value is not None and value > max_value:
013                 print("Invalid input! Please enter a valid value.")
014                 continue
015             return value
016         except ValueError:
017             print("Invalid input! Please enter a number.")
018
019 name = input("Enter patient name: ")
020 age = get_integer("Enter patient age: ", 1, 120)
021
022 while True:
023     phone = input("Enter 10-digit phone number: ")
024     if len(phone) == 10 and phone.isdigit():
025         print("Phone number accepted")
026         break
027     else:
028         print("Invalid phone number! Please enter exactly 10 digits.")
029
030 doctor_price = 0
031 test_total = 0
032 room_total = 0
033 food_total = 0
034
035 while True:
036     print()
037     print("MAIN MENU")
038     print()
039     print("1. Doctor Appointment")
040     print("2. Medical Tests")
041     print("3. Hospital Room")
042     print("4. Food Services")
043     print("5. Generate Bill")
044     print("6. Exit")
045     print()
046
047     choice = get_integer("Enter your choice: ", 1, 6)
048
049     if choice == 1:
050         print()
051         print("DOCTOR APPOINTMENT")
052         print()
053         print("1. General Medicine - Rs. 500")
054         print("2. Cardiology - Rs. 1000")
055         print("3. Orthopedics - Rs. 700")
056         print("4. Dermatology - Rs. 600")
057         print()
058         doctor = get_integer("Enter doctor choice: ", 1, 4)
059         if doctor == 1:
060             doctor_price = 500
061             print("General Medicine selected")
062         elif doctor == 2:
063             doctor_price = 1000
064             print("Cardiology selected")
065         elif doctor == 3:
066             doctor_price = 700
067             print("Orthopedics selected")
068         else:
069             doctor_price = 600
070             print("Dermatology selected")
071
072     elif choice == 2:
073         print()
074         print("MEDICAL TESTS")
075         print()
076         print("1. Blood Test - Rs. 300")
077         print("2. X-Ray - Rs. 500")
078         print("3. ECG - Rs. 400")
079         print("4. MRI - Rs. 2500")
080         print("5. Done")
081         print()
082         while True:
083             test = get_integer("Enter test choice: ", 1, 5)
084             if test == 1:
085                 test_total += 300
086                 print("Blood Test added")
087             elif test == 2:
088                 test_total += 500
089                 print("X-Ray added")
090             elif test == 3:
091                 test_total += 400
092                 print("ECG added")
093             elif test == 4:
094                 test_total += 2500
095                 print("MRI added")
096             else:
097                 print("Medical test selection completed")
098                 break
099
100     elif choice == 3:
101         print()
102         print("HOSPITAL ROOM")
103         print()
104         print("1. General Ward - Rs. 1000 per day")
105         print("2. Semi-Private Room - Rs. 2000 per day")
106         print("3. Private Room - Rs. 3500 per day")
107         print()
108         room = get_integer("Enter room choice: ", 1, 3)
109         days = get_integer("Enter number of days: ", 1)
110         if room == 1:
111             room_price = 1000
112             print("General Ward selected")
113         elif room == 2:
114             room_price = 2000
115             print("Semi-Private Room selected")
116         else:
117             room_price = 3500
118             print("Private Room selected")
119         room_total = room_price * days
120
121     elif choice == 4:
122         print()
123         print("FOOD SERVICES")
124         print()
125         print("1. Breakfast - Rs. 150")
126         print("2. Lunch - Rs. 250")
127         print("3. Dinner - Rs. 250")
128         print("4. Juice - Rs. 100")
129         print("5. Done")
130         print()
131         while True:
132             food = get_integer("Enter food choice: ", 1, 5)
133             if food == 1:
134                 food_total += 150
135                 print("Breakfast added")
136             elif food == 2:
137                 food_total += 250
138                 print("Lunch added")
139             elif food == 3:
140                 food_total += 250
141                 print("Dinner added")
142             elif food == 4:
143                 food_total += 100
144                 print("Juice added")
145             else:
146                 print("Food selection completed")
147                 break
148
149     elif choice == 5:
150         total = doctor_price + test_total + room_total + food_total
151         print()
152         print("HOSPITAL BILL")
153         print()
154         print("Patient Name: ", name)
155         print("Patient Age: ", age)
156         print("Phone Number: ", phone)
157         print()
158         print("Doctor Charges: Rs.", doctor_price)
159         print("Medical Test Charges: Rs.", test_total)
160         print("Room Charges: Rs.", room_total)
161         print("Food Charges: Rs.", food_total)
162         print()
163         print("TOTAL PAYMENT: Rs.", total)
164
165     elif choice == 6:
166         print()
167         print("Thank you for visiting City Care Hospital!")
168         print("Have a nice day!")
169         break
```

Figure 17. Full source-code screenshot of the Hospital Management System implementation.

The source-code screenshot is included to provide a visual record of the complete implementation used for the execution evidence.